SUPERSEDING
SC-A-46624A
24 May 1961

MILITARY SPECIFICATION

TEST SET, INSULATION BREAKDOWN AN/GSM-6( )

1. SCOPE

1.1 This document covers one type of insulation breakdown test set designated as Test Set, Insulation Breakdown, AN/GSM-6( ).

2. APPLICABLE DOCUMENTS

2.1 Documents. The following documents of the issue in effect at date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

T-T-871	Twine, Cotton, Wrapping
FF-B-571	Bolt, Nuts, Studs, and Tap Rivets (and Material For Same)
NN-P-530	Plywood, Flat Panel
QQ-S-781	Steel Castings
MMM-A-260	Adhesive, Water-Resistant, (For Sealing Waterproofed Paper)
PPP-B-621	Box, Wood, Nailed and Lock-Corner
PPP-B-636	Box, Fiberboard
PPP-C-843	Cushioning, Material, Cellulosic
PPP-F-320	Fiberboard, Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes
PPP-P-291	Paperboard, Wrapping, Cushioning
PPP-T-45	Tape, Gummed, Paper, Reinforced and Plain, for Sealing and Securing
PPP-T-97	Tape, Pressure-Sensitive Adhesive, Filament Reinforced

MILITARY

MIL-P-116	Preservation, Methods of
MIL-B-117	Bag and Sleeve, Interior Packaging
MIL-B-131	Barrier Material, Water Vaporproof, Flexible

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MIL-D-3464	Desiccants, Activated, Bagged, Packaging Use and Static Dehumidification
MIL-I-8835	Indicator, Humidity, Card, Chemically Impregnated
MIL-P-11268	Parts, Materials and Processes Used in Electronic Communication Equipment
MIL-G-12803	Gasket Material, Non-Metallic
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes for Ground Signal Equipment

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Table for Inspection by Attributes
MIL-STD-129	Marking for Shipping and Storage
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects
MIL-STD-781	Reliability Tests Exponential Distribution
MIL-STD-810	Environmental Test Methods

DRAWINGS

ELECTRONICS COMMAND

SC-A-46439	List of Accessories for Package Tester
SC-DL-177395	Lead, Test, CX-4698( )/G
DL SM-B-177413	Cable Assembly, Power, Electrical, CX-4696( )/G
DL SM-B-177414	Cable Assembly, Special Purpose CX-4697( )/G
SC-DL-349200	Test Set, Insulation Breakdown, TS-928( )/G

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity, or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies.)

3. REQUIREMENTS

3.1 Description. Test Set, Insulation Breakdown, AN/GSM-6( ) is a portable dielectric test set which provides facilities for measuring dc current at voltages up to 40 kilovolts and provides a high dc breakdown test voltage for testing cables, bushings, transformers and generators for dielectric adequacy.

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3.2 First article samples. The contractor shall furnish two (2) complete first article samples for approval provided the invitation for bid and the contract requires first article samples.

3.3 Construction. The equipment shall be constructed in accordance with the following drawings and the requirements specified herein:

SC-DL-349200
DL SM-B-177413
DL SM-B-177414
SC-DL-177395

3.4 Parts, materials and processes - general. In addition to the requirements of this specification, the requirements of MIL-P-11268, including the selection of requirements therein, shall apply.

3.5 Finish, protective. The equipment shall be finished in accordance with MIL-F-14072. This includes finish of hardware such as screws, etc, and necessary touch-up after mounting. The final paint film of type I surfaces shall be final Film E, semigloss, light gray enamel, conforming to MIL-F-14072.

3.6 Marking. The marking shall conform to MIL-M-13231. Front panel marking shall be Group I as described in that specification.

3.6.1 Visibility. Wherever practicable, parts and assemblies shall be so mounted that their identification markings shall be readily visible with a minimum disassembly of the equipment.

3.6.2 Serial numbers. Serial numbers are required.

3.7 Electrical.

3.7.1 Safety controls. The test set safety controls shall indicate improper polarity and improper grounding of the test set when tested in accordance with 4.7.1.

3.7.2 Voltage, output. The output voltage shall be adjustable from 500 volts to 40,000 volts dc when tested in accordance with 4.7.2.

3.7.3 Leakage current. The leakage current as indicated on the test set ammeter shall be less than 1 microampere for all ammeter ranges when tested in accordance with 4.7.3.

3.7.4 Output and meter calibration.

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3.7.4.1 Current, output. The test set shall be capable of providing the following current output at the loads specified:

3 milliamperes ± 3 percent at 40 kv (See 4.7.4.1.2)
13 milliamperes ± 3 percent at 20 kv (See 4.7.4.2.2)
18 milliamperes ± 3 percent at 10 kv (See 4.7.4.3.2)

3.7.4.2 Accuracy of test set voltmeter. The accuracy of the voltage indications shall be as follows:

When the Standard Reads:	The Test Set shall Read Between:	At Test Set Voltage Range:
40,000 V	38,750 - 41,250 V	50 (See 4.7.4.1.3)
20,000 V	19,500 - 20,500 V	25 (See 4.7.4.2.3)
9,000 V	8,800 - 9,200 V	10 (See 4.7.4.4.1)
5,000 V	4,800 - 5,200 V	10 (See 4.7.4.5.2)
4,000 V	3,900 - 4,100 V	5 (See 4.7.4.6.1)

3.7.4.3 Accuracy of test set ammeter. The accuracy of the current indications shall be as follows:

When the Test Set Reads:	At Test Set Ammeter Range:	The Standard Shall Read Between:
18 mA	1000 (See 4.7.4.4.2)	16.5 - 19.4 mA
3 mA	100 (See 4.7.4.1.4)	2.92 - 3.10m A
500 μA	10 (See 4.7.4.7.2.1)	486 - 514 μA
50 μA	1 (See 4.7.4.7.3)	48.63 - 51.38 μA.

When the Standard Reads:	At Test Set Ammeter Range:	The Test Set Shall Read Between:
500 μA	100 (See 4.7.4.7.2.2)	363 - 638 μA

3.7.5 Polarity. The equipment shall be capable of providing either negative or positive polarity with reference to ground at the output terminal when tested in accordance with 4.7.5.

3.7.6 Ripple. The ripple in the output voltage shall be less than one percent with a 5 megohm resistive load (see 4.7.6).

3.7.7 Internal resistance. The internal resistance of the rectifier unit shall be 0.7 megohm ± 20 percent. (See 4.7.7)

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3.7.8 Overload and meter protective circuits.

3.7.8.1 Current. The current indicated by the test set ammeter shall be 15 mA ± 20 percent when tested in accordance with 4.7.8.1.

3.7.8.2 Spark gap. An arc shall occur between the jumper wire and case at a point at which the ammeter pointer does not exceed full scale. Also, the spark gap electrodes, E-102, shall glow when breakdown occurs. (See 4.7.8.2).

3.8 Service conditions. The equipment and assemblies shall be constructed so that no fixed part shall become loose, cracked, flaked, pitted, or blistered, no movable part or control shall be shifted in setting, position or adjustment, and no permanent degradation shall be caused in performance below that specified in the following service conditions.

3.8.1 Low temperature. The equipment and assemblies shall be subjected to the low temperature extremes of 40°F operating and -65°F storage and shall meet the electrical requirements of 3.7.4.1 at times specified when tested in accordance with 4.8.1.

3.8.2 High temperature. The equipment and assemblies shall be subjected to the high temperature extremes of 125°F operating and 160°F storage and shall meet the electrical requirements of 3.7.4.1 at times specified when tested in accordance with 4.8.2.

3.8.3 Altitude. The equipment and assemblies shall be subjected to the altitudes specified and shall meet the electrical requirements of 3.7.4.1 at times specified when tested in accordance with 4.8.3.

3.8.4 Fungus. The equipment and assemblies shall show no evidence of viable fungus and shall meet the requirements of 3.7.4.1 at the end of the incubation period when tested in accordance with 4.8.4.

3.9 Visual and mechanical. The equipment and assemblies shall meet the requirements of MIL-STD-252 as specified in 4.9.

3.10 Interchangeability. Like units, assemblies, subassemblies, and replaceable parts shall be physically and functionally interchangeable without modification of such items or the equipment. Individual items shall not be hand picked for fit or performance. Reliance shall not be placed on any unspecified dimension, rating, characteristics, etc. (See 4.10).

3.11 Bounce preconditioning. Each Test Set AN/GSM-6( ) produced shall be subjected to the bounce preconditioning test of 4.6 prior to Group A inspection. After bounce preconditioning the Test Sets, AN/GSM-6( ) shall not be repaired, aligned, cleaned, or otherwise changed prior to performing electrical preconditioning and Group A acceptance test.

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3.12 Electrical preconditioning. Each Test Set AN/GSM-6( ) shall be subjected to the electrical preconditioning test of 4.11 after bounce preconditioning and prior to Group A inspection. The equipment shall be monitored for 24 hours during period of test. At the conclusion of electrical preconditioning test (4.11), the test set shall meet Group A inspection requirements.

3.13 Workmanship. The equipment and assemblies shall be manufactured and assembled in accordance with the applicable portions of MIL-P-11268 as follows:

General requirements for plastic material and plastic parts.
Wiring and cabling, including:
- Slack.
- Protection.
- Clearance.
- Splicing.
- Connections, general.
- Grounding, general.
- Shielding on wire and cable.

Flux and cleaning agents for soldering.
Process for soldering.
Cleaning.
Riveting.
General requirements for securing of parts.
Cleaning of welding.
Process for welding.
Mechanical operations of controls.
Gears.
Self-tapping screws.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the contract and specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection. Inspection shall be classified as follows:

(a) First article inspection (does not include preparation for delivery) (see 4.3).

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(b) Inspection covered by subsidiary documents (see 4.4).

(c) Quality conformance inspection.

(1) Quality conformance inspection of equipment before preparation for delivery (see 4.5).

(2) Quality conformance inspection of preparation for delivery (see 4.12).

4.3 First article inspection. This inspection shall be performed by the contractor and witnessed by a Government representative unless otherwise specified in the contract. It shall consist of Group A, B and c inspections (see Tables I, II and III, respectively) on the two (2) samples of the completely assembled Test Set AN/GSM-6( ). The first article samples shall be inspected in this order: Group A and B for all units; and designated units for specific Group C inspections. Designation of units to specific Group C inspections shall be determined by the Government production engineer. After completion of all Group C tests, conforming units must be reinspected and pass all Group A inspections.

4.4 Inspection covered by subsidiary documents. The following shall be inspected under the applicable subsidiary documents as part of the inspection of equipment before preparation for delivery.

Item	Where required
Finish	3.5
Marking	3.6
Workmanship	3.13

4.5 Quality conformance inspection of equipment before preparation for delivery. The contractor shall perform the inspection specified in 4.1 and 4.5.1 through 4.5.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition, the Government--at its discretion may perform all or any part of the specified inspection, to verify the contractor's compliance with specified requirements. Test equipment for Government verification inspection shall be made available by the contractor.

4.5.1 Group A inspection. Each unit produced shall be inspected by the contractor for conformance to all the Group A requirements and tests in Table I. Each lot shall be subject to Government verification and this inspection, including sampling, shall conform to Table I and the ordinary inspection procedures of MIL-STD-105.

4.5.1.1 Order of inspection within Group A. Group A inspection shall be performed in any order which is satisfactory to the Government.

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Table I - Group A inspection

Inspection	Req. Para.	Insp. Para	AQL	
			Major	Minor
Visual and mechanical	3.9	4.9		
Safety controls	3.7.1	4.7.1		
Voltage, output	3.7.2	4.7.2		
Leakage, current	3.7.3	4.7.3		
Output and meter calibration	3.7.4	4.7.4		
Polarity	3.7.5	4.7.5		
Ripple	3.7.6	4.7.6		
Internal resistance	3.7.7	4.7.7		
Overload and meter protective circuits	3.7.8	4.7.8		

4.5.2 Group B inspection. This inspection, including sampling, shall conform to Table II and to the special procedures for small sample inspection of MIL-STD-105. Group B inspection shall be performed on inspection lots that have passed Group A inspection, and on samples selected from units that have been subjected to and met the Group A inspection.

4.5.2.1 Order of inspection within Group B. Group B inspection shall be performed in any order which is satisfactory to the Government.

Table II - Group B inspection

Inspection	Req. Para.	Test Para.	AQL
Interchangeability	3.10	4.10	

4.5.3 Group C inspection. This inspection shall consist of the inspection specified in Table III and shall be performed on sample units that have been subjected to and met Group A and B inspection. Sample units shall be selected in accordance with 4.5.3.1.

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4.5.3.1 Sampling for inspection. Two (2) samples shall be selected at random, one (1) sample for each of Group I and Group II in Table III from every fifty (50) units and fraction thereof produced. The first samples selected shall be at the start of the contract from the first quality conformance inspection production lot.

4.5.3.1.1 Thereafter, two (2) samples of complete equipment shall be selected at random, one (1) sample for each Group I and Group II in Table III from every fifty (50) units and fraction thereof produced.

4.5.3.2 Evaluation of samples submitted to individual Group C inspection. After referenced samples in 4.5.3.1 have been subjected to an individual Group C inspection test and prior to subjection to the next Group C test (unless otherwise specified in the individual test) the sample shall meet the requirements of 3.7.4.1.

4.5.3.3 Noncompliance. The contractor shall immediately report in writing each Group C failure occurrence, including details of the failure and characteristics affected. The contractor shall immediately investigate the cause of failure and further report the results of investigation and details of the proposed corrective action on (i) the process and materials, as applicable, and (ii) all units of product which were manufactured under the same conditions and which the Government considers subject to the same failure. Reports shall be forwarded to the responsible technical activity designated in the contract through the Quality Assurance Representative. After corrective action has been taken, additional sample units shall be subjected to Group C inspection (all inspections, or the inspections which the sample failed, at the option of the Government). The final acceptance and shipment will be withheld until the Group C reinspection results have shown that the corrective action was effective.

4.5.3.4 Order of inspection within Group C. Group C inspection shall be performed in any order which is satisfactory to the Government.

Table III - Group C inspection

Inspection	Req. Para.	Test Para.
Group I		
Low temperature	3.8.1	4.8.1
High temperature	3.8.2	4.8.2
Group II		
Altitude	3.8.3	4.8.3
Fungus	3.8.4	4.8.4

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4.5.4 Reinspection of conforming Group B and C sample units. Unless otherwise specified, sample units which have been subjected to and passed both Group B and Group C inspection may be accepted on contract or order provided all damage is repaired and the sample units are subjected to and pass Group A inspection.

4.5.5 Disposition of Group C sample units. Unless otherwise specified, sample units which have been subjected to and passed Group C inspection may be delivered on the contract or order provided units meet requirements of 4.5.4.

4.6 Bounce preconditioning. Each Test Set, AN/GSM-6( ) produced shall be placed in its normal operating position on the table of a package tester of suitable capacity as made by the L.A.B. Corporation, Skaneateles, New York, with accessories as per Model 1000SC and SC-A-46439, or equal. The package tester, shafts in phase, shall have a speed such that it is just possible to insert a 1/32 inch thick strip of material under one corner or edge of the equipment under test to a distance of 3 inches as the equipment bounces. The equipment shall be subjected to this preconditioning for 1 minute and shall meet the requirements of 3.11.

4.7 Electrical. Preliminary: adjust the spark gap electrodes for a clearance of 3 to 4 thousands of an inch and lock in position.

Table IV - Test Equipment Description

TE-1:	Electrostatic Voltmeter, Model ESH as made by Sensitive Research Instrument Corporation, or equal. Ranges: 0-5 kv, 0-10 kv, 0-20 kv, 0-30 kv Accuracy: ±1 percent of full scale.
TE-2:	Electrostatic Voltmeter, Model ESH as made by Sensitive Research Instrument Corporation, or equal. Range: 0-50 kv Accuracy: 1/2 percent of full scale.
TE-3:	A. C. Vacuum Tube Voltmeter, Model 300G as made by Ballantine Laboratories Incorporated, or equal. Ranges: (6) 0.01 - 0.1 -1 - 10 - 100 - 1000 full scale. Accuracy: ± 1 percent at all points on scale.

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Table IV - Test Equipment Description (Cont'd)

TE-4: Variable Auto Transformer, Model W5MT3A as made by General Radio Company, or equal.

Range: 0-135 volts, 50-60 cycles, 0-5 amps.

TE-5: Load Resistance Box as shown in Figure 6.

TE-6: D. C. Ammeter as made by Industrial Instruments Incorporated (per Drawing 4285-294), or equal (see Figure 7).

Ranges: 0-50 microamps, 0-60 microamps, 0-600 microamps, 0-5 milliamps, 0-15 milliamps, 0-20 milliamps.
Accuracy: ± 0.25 percent.

TE-7: Low Resistance Ohmmeter, Model LRO-1 as made by Industrial Instrument Incorporated, or equal.

Ranges: 0-10, 0-1, 0-.1 ohms full scale.
Accuracy: ± 3 percent.

TE-8: Multiplier, Model 640 as made by Triplett Electric Instruments Company, or equal.

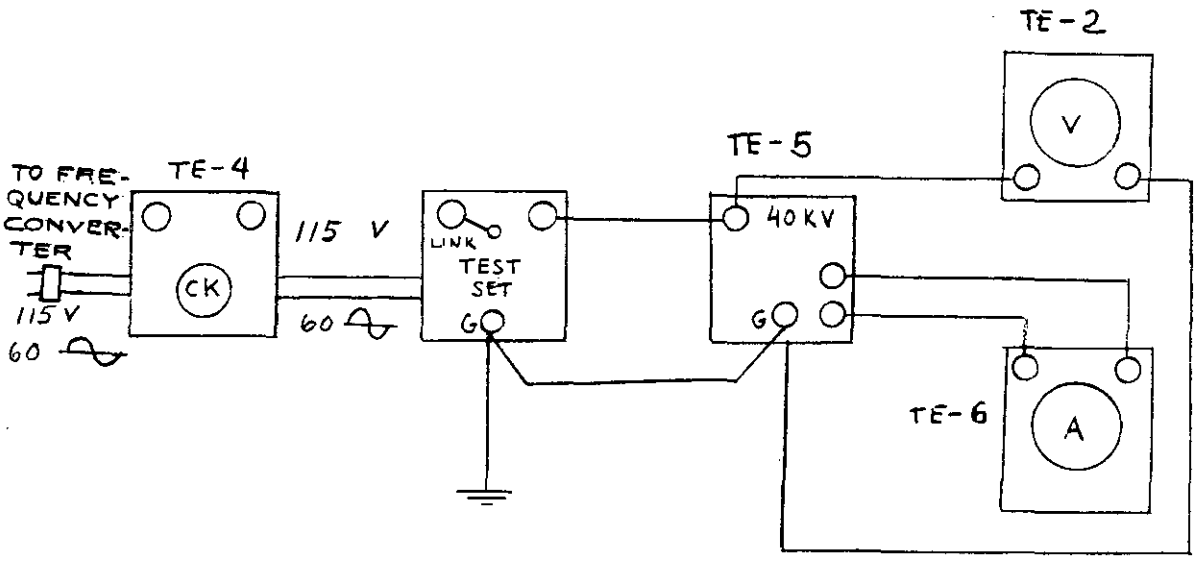
TE-9: Megohmmeter, Model L7 as made by Industrial Instruments Incorporated, or equal.

Accuracy: ± 5 percent.

4.7.1 Safety controls.

4.7.1.1 Preparation. The line frequency shall be 60 hertz with the test set powered through the Line Voltage Control, TE-4 (see Table IV). (CAUTION: Make sure that the test set is grounded directly and separately. Ground the test set high voltage terminals before making connections to test equipment.) Connect the link between the guard and either one of the high voltage terminals of the test set, the polarity shall be that of the non-linked terminal. Actuate the Test Set Meter Polarity Switches to the appropriate position. Rotate the Test Set Meter Range Switches to their highest range. Operate the Test Set Circuit Breaker to the OFF position. Rotate the Test Set Voltage Control to its full counter-clockwise position. Connect the Input Cable to the Voltage Control, TE-4. Set the Voltage Control, TE-4, for 115 volts.

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SEE TABLE IV

FIGURE 1

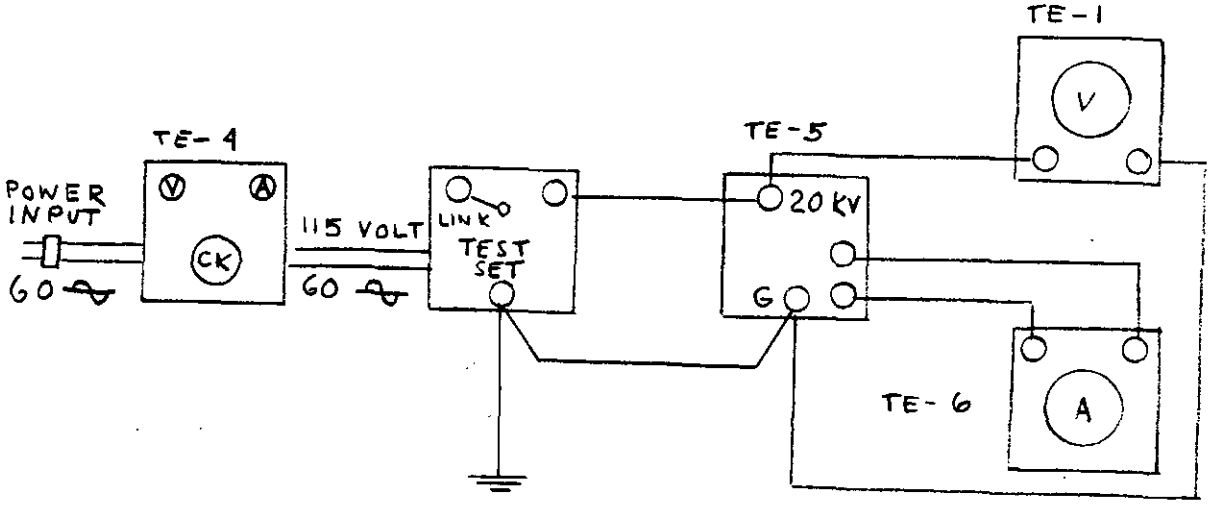


FIGURE 2

SEE TABLE IV

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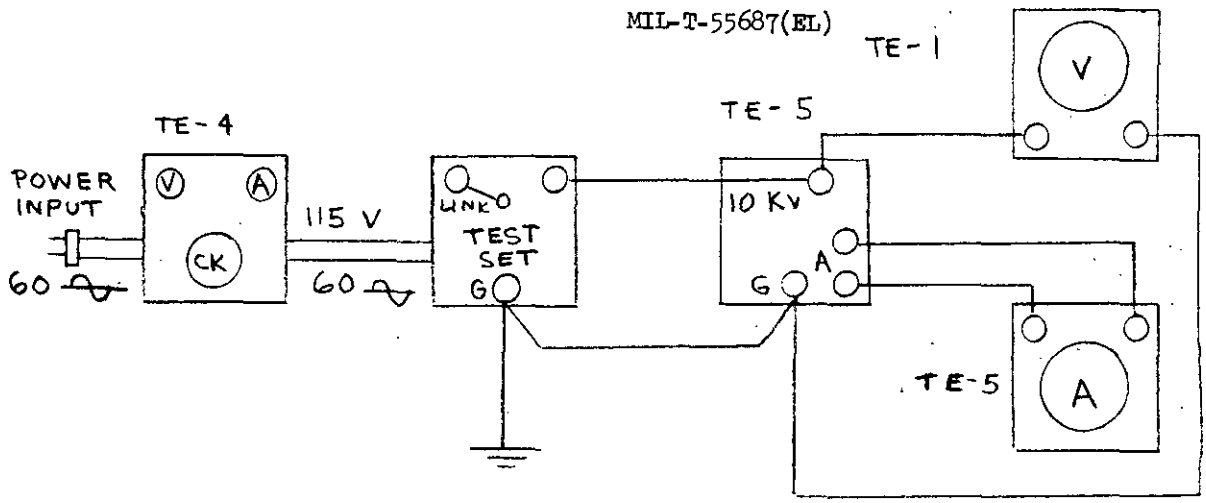
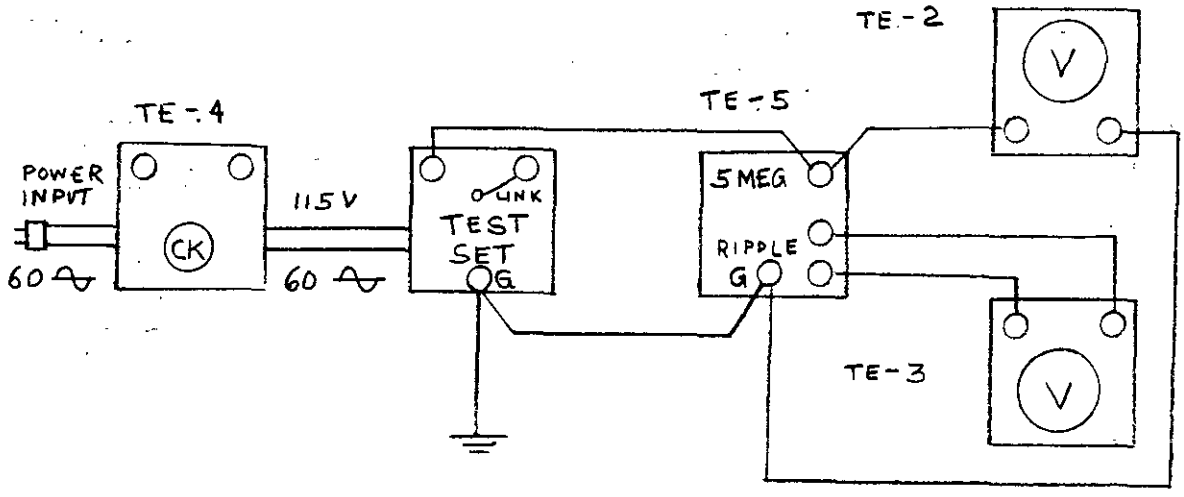


FIGURE 3

SEE TABLE IV



SEE TABLE IV

FIGURE 4

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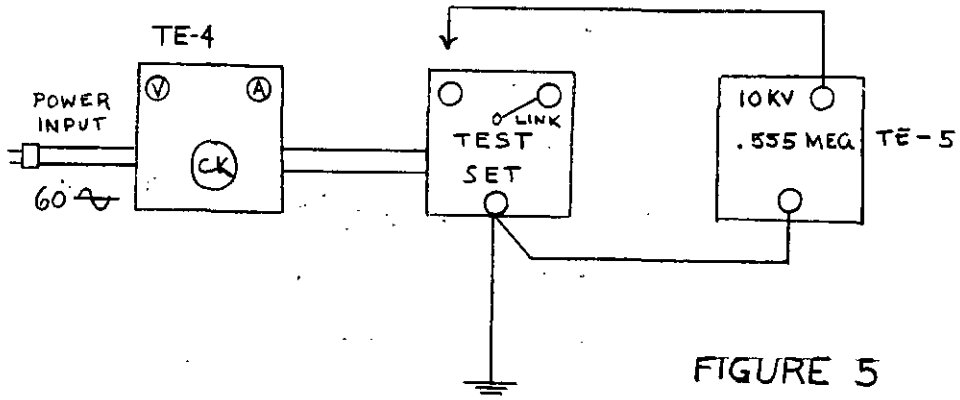


FIGURE 5

SEE TABLE IV

HIGH VOLTAGE RESISTANCE LOAD BOX IE-5

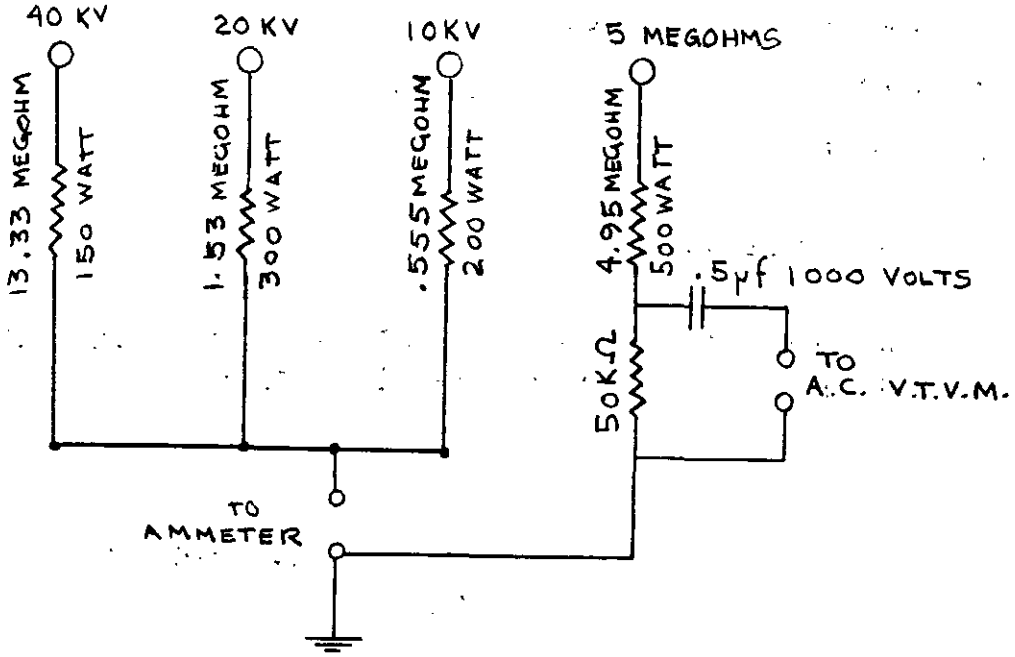


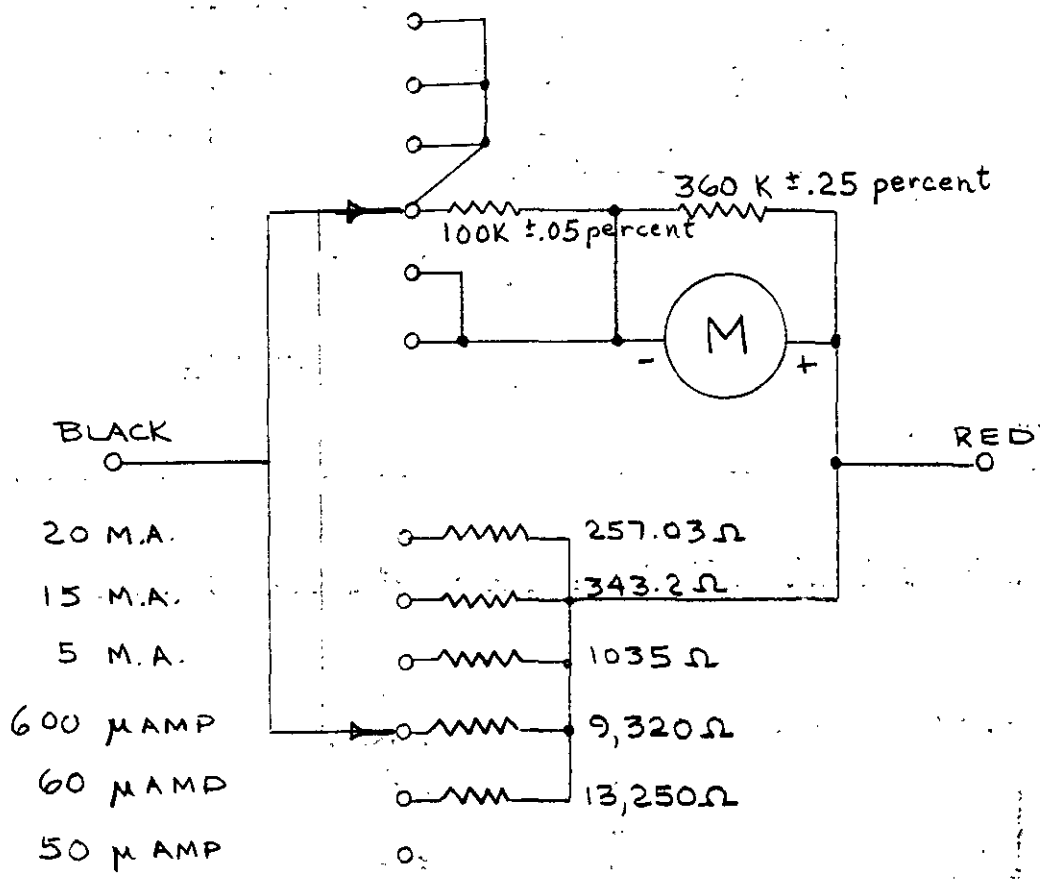
FIGURE 6

SEE TABLE IV

RESISTANCE TOLERANCE $^{+0}_{-1\%}$

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6 RANGE AMMETER TE-6



SEE TABLE V

ALL RESISTORS ADJUSTED FOR A FULL SCALE CURRENT ACCURACY OF ± 0.2%.

FIGURE 7

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4.7.1.2 Safety controls test. Operate the Test Set Circuit Breaker to ON. The green light should go ON. If the polarity lamp (white) lights, reverse the polarity of the input cable plug; if the polarity lamp did not light when the circuit breaker was first turned ON reverse the polarity of the input cable plug and see if the polarity lamp lights. Reverse the polarity of the input cable plug. Turn the test set circuit breaker to the OFF position. Remove the ground wire from the case. Actuate the circuit breaker to ON, the white light shall go ON. Turn the test set circuit breaker OFF and re-connect the ground wire to the case. (See 3.7.1)

4.7.2 Voltage output. With the test set voltage control turned fully counter-clockwise, the output voltage read on the test set voltmeter shall be less than 500 volts. The test set voltage control shall be rotated to its full clockwise position, the test set voltmeter shall read at least 40,000 volts. (See 3.7.2.)

4.7.3 Leakage current. The test set ammeter range switch shall be rotated through each range to the lowest range and the readings shall meet the requirements of 3.7.3.

4.7.4 Output and meter calibration.

4.7.4.1 40 kv measurements.

4.7.4.1.1 Preparation. (see Figure 1). The test set voltage control shall be in its full counter-clockwise position and the circuit breaker shall be in the OFF position. Connect the high voltage terminal of TE-2 and the terminal marked 40 kilovolts of TE-5 to the non-linked terminal of the test set. Connect TE-6 to the ammeter terminals of TE-5.

4.7.4.1.2 Output at 40 kv. Turn the test set ammeter range switch to 100. Turn the standard ammeter range switch of TE-6 to 5 mA. Turn the test set voltmeter range switch to 50. Actuate the test set circuit breaker to ON, depress the output button, rotate the test set voltage control knob so that the standard voltmeter, TE-2 reads exactly 40 kv. The current indicated on the standard ammeter, TE-6, shall read 3 mA $\pm$ 3 percent (see 3.7.4.1).

4.7.41.3 Accuracy of test set voltmeter at 40 kv. The voltage indicated by the test set voltmeter shall be between 38,750 and 41,250 volts (see 3.7.4.2).

4.7.4.1.4 Accuracy of test set ammeter at 3 mA. Rotate the voltage control knob of the test set in order to read exactly 3 mA on the test set ammeter. The current indicated on the standard ammeter shall be between 2.92 and 3.10 mA. (See 3.7.4.3)

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4.7.4.2 20 kv measurements.

4.7.4.2.1 Preparation (see Figure 2). The test set voltage control shall be in its full counter-clockwise position and the circuit breaker shall be in the OFF position. Disconnect the wire from the 40 kv terminal of the load box and connect it to the 20 kv terminal. Connect TE-1 between the test set ground and the 20 kv terminal of Load Box, TE-5.

4.7.4.2.2 Output at 20 kv. Turn the test set ammeter range switch to 1000. Turn the test set voltmeter range switch to 25. Turn the standard ammeter, TE-6 range switch to 15 mA set TE-1 range control for 20 kv. Actuate the test circuit breaker to ON, depress the output control. Rotate the test set voltage control knob so that the standard voltmeter. TE-1 reads exactly 20 kv. The current indicated on the standard ammeter, TE-6, shall be 13 mA ± 3 percent. (See 3.7.4.1)

4.7.4.2.3 Accuracy of test set voltmeter at 20 kv. The voltage indicated on the test set voltmeter shall be between 19,500 and 20,500 volts. (See 3.7.4.2)

4.7.4.3 10 kv measurements.

4.7.4.3.1 Preparation for 10 kv measurements (see Figure 3). The test set voltage control shall be in its full counter-clockwise position and the circuit breaker shall be in the off position. Disconnect the wires from the 20 kv terminal of the load box, TE-5 and connect them to the 10 kv terminal.

4.7.4.3.2 Output at 10 kv. Turn the test set ammeter range switch to 10. Change the standard voltmeter, TE-1, range control to 10 kv. Turn the standard ammeter, TE-6, range switch to 20 mA. Actuate the test set circuit breaker to ON. Depress the output control button. Rotate the test set voltage control knob so that the standard voltmeter, TE-1, reads exactly 10 kv. The current indicated on the standard ammeter shall be 18 mA ± 3 percent. (See 3.7.4.1)

4.7.4.3.3 Repeat paragraph 4.7.4.3.2. for a line frequency of 50 hertz.

4.7.4.4 9 kv measurements.

4.7.4.4.1 Accuracy of test set voltmeter at 9 kv. Return the line frequency to 60 hertz. Rotate the test set voltage control knob so that the standard voltmeter TE-1 reads exactly 9 kv. The voltage indicated on the test set voltmeter shall be between 8,800 and 9,200 volts. (See 3.7.4.2)

4.7.4.4.2 Accuracy of test set ammeter at 18 mA. Rotate the test set voltage control knob so that the test set ammeter reads exactly 18 mA. The current indicated on the standard ammeter, TE-6 shall be between 16.5 and 19.4 mA. (See 3.7.4.3)

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4.7.4.5 5 kv measurements.

4.7.4.5.1 Preparation for 5 kv measurements. The test set voltage control knob shall be in its full counter-clockwise position and the circuit breaker shall be in the OFF position. Set the range control of the standard voltmeter, TE-1, to 5 kv.

4.7.4.5.2 Accuracy of test set voltmeter at 5 kv. Actuate the test set circuit breaker to ON. Depress the output control button. Rotate the voltage control knob so that the standard voltmeter TE-1, reads exactly 5 kv. The voltage indicated on the test set voltmeter shall be between 4,800 and 5,200 volts. (See 3.7.4.2)

4.7.4.6 4 kv measurements.

4.7.4.6.1 Accuracy of test set voltmeter at 4 kv. Rotate the test set voltage control to its full counter-clockwise position. Rotate the test set voltmeter range switch to 5. Rotate the test set voltage control so that the standard voltmeter TE-1 reads exactly 4000 volts. The voltage indicated on the test set voltmeter shall be between 3900 and 4100 volts. (See 3.7.4.2)

4.7.4.7 Accuracy of test set ammeter.

4.7.4.7.1 Preparation. With the test set voltage control at its full counter-clockwise position and the circuit breaker in the OFF position, disconnect the standard voltmeter, TE-1. Rotate the test set voltmeter range switch to 10. Rotate the test set ammeter range switch to 10. Rotate the standard ammeter range switch to 600 microamps. Disconnect the wire from the 10 kv terminal of the load box and connect it to the 40 kv terminal.

4.7.4.7.2 Accuracy of test set ammeter at 500μamps.

4.7.4.7.2.1 Test set ammeter range switch at 10. Rotate the test set voltage control knob so that the test set ammeter reads exactly 500μ amps. The current indicated by the standard ammeter shall be 486 to 514μamps. (See 3.7.4.3)

4.7.4.7.2.2 Test set ammeter range switch at 100. The current indicated on the test set ammeter shall be between 363 and 638μamps. (See 3.7.4.3)

4.7.4.7.3 Accuracy of test set ammeter at 50μamps. Rotate the test set voltage control knob to its full counter-clockwise position. Rotate the test set ammeter range switch to 1. Rotate the standard ammeter, TE-6, range switch to 60μamps. Rotate the test set voltage control knob until the test set ammeter reads exactly 50μamps. The current indicated by the standard ammeter shall be between 48.63 and 51.38μamps. (See 3.7.4.3) Rotate the test set voltage control to its full counter-clockwise position and actuate the circuit breaker to the OFF position.

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4.7.5 Polarity. With the test set voltage control knob at its full counter-clockwise position, the circuit breaker at the OFF position, and all test equipment except standard ammeter TE-6 disconnected, remove the link from one high voltage terminal of the test set and connect it to the other one. Reverse the position of the test set meter polarity switches and reverse the position of the wires to the standard ammeter, TE-6. Repeat the procedures and tests of paragraphs 4.7.4.1.1, 4.7.4.1.2. Rotate the test set voltage control to its full counter-clockwise position and operate the circuit breaker to the OFF position.

4.7.6 Ripple.

4.7.6.1 Preparation. Connect the non-linked high voltage terminal of the test set to the 5 meg terminal of the load box, TE-5. Connect the high voltage terminal of the electrostatic voltmeter, TE-2, to the non-linked terminal of the test set. Connect the low or ground terminals of the load box, TE-5 and the electrostatic voltmeter, TE-2, to the ground terminal of the test set. Connect the A.C. vacuum tube voltmeter, TE-3, to the load box terminals marked "RIPPLE". (See figure 4) Set the V.T.V.M., TE-3, range switch to the 10 volt range. Rotate the test set ammeter range switch to 1000.

4.7.6.2 Ripple measurement. Actuate the test set circuit breaker to the ON position. Rotate the test set voltage control to its full clockwise rotation. The reading of TE-3 shall be 1/10,000th of the reading of TE-2, or less. (See 3.7.6) Rotate the test set voltage control knob fully counter-clockwise. Operate the circuit breaker to the OFF position and disconnect all test equipment.

4.7.7 Internal resistance.

4.7.7.1 Preparation (see Figure 5). Set the test set voltmeter range switch to 5, and the ammeter range switch to 100. Actuate the test set circuit breaker to the ON position. Rotate the test set voltage control until the voltmeter reads exactly 5 kilovolts. This value (5 kilovolts) shall be E1. (Make sure that the line input voltage is exactly 115 volts.) Rotate the line voltage knob of TE-4 fully counter-clockwise. Operate the test set circuit breaker to the OFF position. Connect the non-linked terminal of the test set to the 10 kv (.555 megohm) terminal of the load resistance box, TE-5, and the ground terminal of TE-5 to the case ground of the test set. Actuate the test set circuit breaker to ON.

4.7.7.2 Internal resistance, measurements. Rotate the line voltage control knob of TE-4 while holding down the test set output voltage button until the line voltage is exactly 115 volts. Read the voltage, E2, and the current, I, indicated on the test set voltmeter and ammeter. The internal resistance of the rectifier, R is:

$$R = \frac{E1 - E2}{I}$$

The internal resistance value, R, shall meet the requirements of 3.7.7.

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4.7.8 Overload and meter protective circuits.

4.7.8.1 Current. With the test set voltage control knob in its full counter-clockwise position, the circuit breaker in the OFF position, and all test equipment disconnected, use a short piece of #14 (or large #) insulated copper wire to connect the non-linked high voltage terminal to the case ground of the test set. Set the test set voltmeter range switch to 50 kv and the ammeter range switch to 1000. Actuate the test set circuit breaker to the ON position. Depress the output voltage button and slowly rotate the voltage control knob until the overload lamp lights. The current indicated by the ammeter shall be 15 mA ± 20 percent.

4.7.8.2 Spark gap. Rotate the test set voltage control knob fully counter-clockwise and actuate the circuit breaker to the OFF position. Remove the jumper wire from the case ground of the test set and adjust it by bending so that the bare end is 1/2 inch away from the case ground terminal. Rotate the test set ammeter range switch to 10. Actuate the circuit breaker to the ON position. Depress the output voltage button and slowly rotate the voltage control knob until a spark occurs between the bare end of the jumper wire and the case. The test set shall meet the requirements of 3.7.8.2 when the voltage control knob is rotated back and forth in order to obtain a series of sparks (not a continuous arc). Rotate the test set voltage control knob to its full counter-clockwise position and actuate the circuit breaker to the OFF position.

4.8 Service conditions. The equipment and assemblies shall be tested in accordance with the following service conditions.

4.8.1 Low temperature. The equipment and assemblies in their normal operating configuration shall be subjected to the low temperature test specified in Procedure I, Method 502 of MIL-STD-810, with the temperature in Step 2 held for twenty-four (24) hours and the temperature in Step 4 held for four (4) hours, and shall meet the requirements of 3.8.1.

4.8.2 High temperature. The equipment and assemblies in their normal operating configuration shall be subjected to the high temperature test specified in Procedure I, Method 501 of MIL-STD-810, with the temperature in Step 4 held for four (4) hours, and shall meet the requirements of 3.8.2. Exposure of the equipment and assemblies at the high, non-operating temperature extreme shall be for four (4) hours, not the forty-eight (48) hours as stated in Step 3 of Procedure I.

4.8.3 Altitude. The equipment and assemblies in their normal operating configuration shall be subjected to the altitude test specified in Procedure I, Method 500 of MIL-STD-810, with altitude in Step 3 held for two (2) hours. (See 3.8.3)

4.8.4 Fungus. The equipment and accessories in their normal operating configuration shall be subjected to 28 days of the fungus test specified in Procedure I, Method 508 of MIL-STD-810 and shall meet the requirements of 3.8.4.

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4.9 Visual and mechanical. The equipment and assemblies shall be examined for the defects listed in MIL-STD-252 and shall meet the requirements specified in 3.9.

4.10 Interchangeability. The dimensions listed below shall be measured to determine conformance to the physical interchangeability of 3.10. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

(a) External and internal dimensions of covers and insertable assemblies, when such dimensions affect mating of parts.

(b) Dimensions of cavities, when such dimensions affect insertion of items.

(c) Location of hinges and fasteners on separable parts or assemblies which must mate, such as covers and mountings.

(d) Location of connectors, locking pins, fasteners, slides, and mountings which receive mating parts of plug-in assemblies and major units; and location of the mating parts on the plug-in assembly or major unit.

4.11 Electrical preconditioning. Each Test Set, AN/GSM-6( ) produced shall be placed in its normal operating position on a test table with room temperature at 77°F ± 5°F, with an operating voltage of 115 Vac applied from a suitable power source. With all controls in the normal operating position, the equipment shall be turned to a voltage of 20,000 V on the highest range and remain on for a continuous 24 hour period. In case of parts failure, the equipment shall be repaired, recalibrated, and the electrical preconditioning test repeated and shall meet the requirements of 3.12. A failed item analysis report shall be submitted on each part failure with contents of report per 5.10.5 of MIL-STD-781.

4.12 Quality conformance inspection of preparation for delivery. Preparation for delivery shall be inspected in accordance with MIL-P-116 to determine conformance to the requirements of Section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging of Insulation Breakdown AN/GSM-6( ), shall be Level A or C as specified. (See Section 6.2(b))

5.1.1 Level A.

5.1.1.1 Cleaning. Cleaning shall be accomplished in accordance with process C-1, of MIL-P-116.

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5.1.1.2 Drying. Drying shall be accomplished in accordance with the applicable procedures of MIL-P-116.

5.1.1.3 Preservation application. None required.

5.1.1.4 Unit packaging. AN/GSM-6( ) shall be packaged in accordance with items 1 through 19 of Figure 8 and Table V.

5.1.2 Level C. AN/GSM-6( ) shall be packaged in a manner that will afford adequate protection against corrosion, deterioration, and damage during shipment from the supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B or C, as specified. (See 6.2(b))

5.2.1 Level A. AN/GSM-6( ) shall be packed in accordance with items 20 through 22 of Figure 8 and Table V.

5.2.2 Level B. AN/GSM-6( ), shall be packed as specified in 5.2.1.

5.2.3 Level C. AN/GSM-6( ), shall be packed in shipping containers in a manner that will afford adequate protection against damage to the package and its contents during shipment from the supply source to the first receiving activity. Shipping containers shall comply with the rules and regulations of the common carrier as applicable to the mode of transportation.

5.3 Marking. In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The test set is used for testing specimens such as insulating materials, insulating structures, and the insulation of a complete electric component. The insulation of the test specimens is subjected to high dc voltages, comparable to those that would be impressed on the test specimen under operating conditions.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this exhibit and any amendment thereto.

(b) Level of packaging and level of packing required for shipment (Levels A, B or C) (see Section 5).

(c) If rough handling and functional tests are required.

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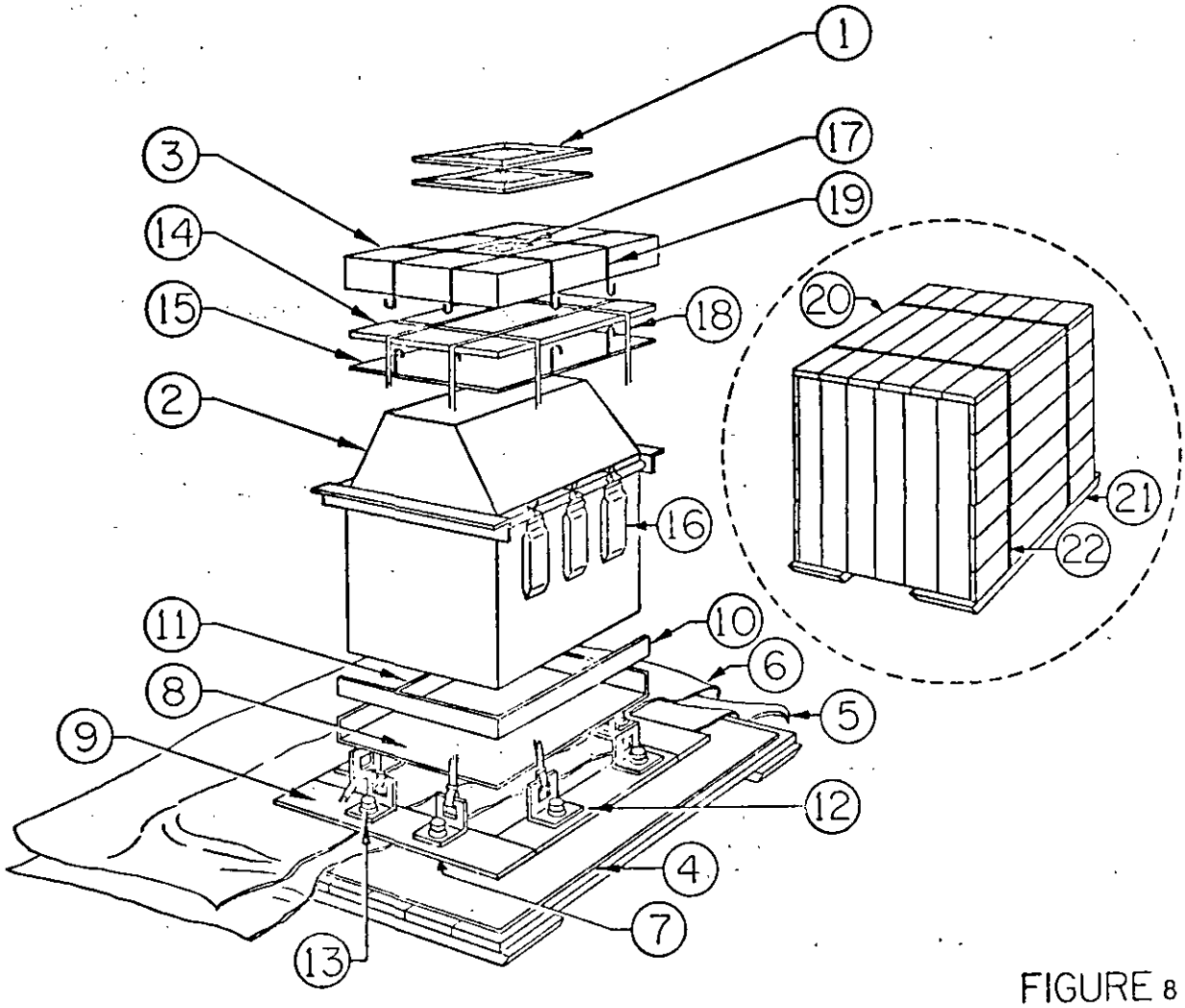


FIGURE 8

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BILL OF MATERIAL
TABLE V

REF No.	ITEM Nomenclature	No. Reqd.	Method MIL-P-116	SIZE - INCHES L	W	D	REQUIREMENT Spec No.	T	C	V	G	F	S
1	Technical Manual	2	IC-1										
	Bag	2		9	11 1/2		MIL-B-117	II	C				
2	Test Set AN/GSM-6( )	1	IIA	20	13 3/8	20 1/2	Bare weight-	134 lbs					
2.1	Lamps	3	III										
	Wrap	3		1	3	1/4	PPP-C-843	III	B				
	Bag	3		3	5		MIL-B-117	II	C				
2.2	Cables	3	III										
	Twine	9		12			T-T-871	I					
3	Box, Fiber	1		13 3/4	13 1/4	4 1/4	PPP-B-636	CF	DOM	SW	200		RS
	Tape	2		20	3		PPP-T-45						
				Items 2.1 and 2.2 shall be placed in				Item 3.					
4	Pad	1		25 1/2	23 1/4	1/8	PPP-F-320	DF	WR	SW	V3c		
5	Barrier	1		100	41		MIL-B-131		1				
6	Paperboard Wrapping	1		90	36		PPP-P-291	II	2				
7	Gaskets	16					MIL-G-12803	III	2		4		
	Adhesive						MMM-A-260						
8	Pad	1		18	18	1/8	PPP-F-320	CF	WR	SW	V3c		
9	Wood Blocking Frame	1	OD	17 3/4	17 1/4	1	PPP-B-621						
9.1	Block, Wood	2		17 3/4	3	1							
9.2	Block, Wood	2		11 1/4	3	1							
10	Pad	2		17 3/4	2	1/8	PPP-F-320	CF	WR	SW	V3c		
11	Pad	2		11 1/4	2	1/8	PPP-F-320	CF	WR	SW	V3c		
12	Angle Steel	8		2	2	1/8							
	Angles shall be formed from 2" x 2" x 1/4" steel angle, drill to receive 3/8" dia bolts and slotted to receive 3/4" metal strapping.												
13	Bolts w/nuts washers	8		4	3/8 Dia		FF-B-571						
14	Plywood	1		14 3/8	13 1/2	3/4	NN-P-530						
15	Pad	1		14 3/8	13 1/2	1/8	PPP-F-320	CF	WR	SW	V3c	C	
16	Desiccant	128	Units				MIL-D-3464						
17	Humidity Indicator	1					MIL-I-8835						
18	Strapping	4		68	3/4	.020	QQ-S-781	I	B		2		

Lev. A - PACKED WEIGHT
Lev. A - PACKED VOLUME

T - Type V - Variety F - Flute
C - Class G - Grade S - Style

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BILL OF MATERIAL (Continued)
TABLE V

REF No.	ITEM Nomenclature	No. Reqd.	Method MIL-P-116	SIZE - INCHES L	W	D	REQUIREMENT Spec No.	T	C	V	G	F	S
19	Banding	4		50	1/2		PPP-T-97						
20	Box, Wood	1		24	21 3/4	28 1/4	PPP-B-621						7
21	Strapping	2		128	3/4	.020	QQ-S-781	I	B		2		

Lev. A - PACKED WEIGHT
Lev. A - PACKED VOLUME

T - Type
C - Class
V - Variety
G - Grade
F - Flute
S - Style

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(d) Quantity of first articles required.

(e) Place of final inspection.

(f) Technical literature required.

(g) Quantity of tools and running spare parts required.

(h) Marking for shipment and shipping containers.

(i) Dimensional data. Dimensions of packaging materials in the Bill of Materials, Table VI, are based on the dimensions of the equipment cited in the table. When the dimensions of the items vary, the dimensions of the packaging material shall be adjusted accordingly.

6.3 Nomenclature and type designation. The contractor should apply for nomenclature and type designation in accordance with the applicable clause in the contract. The parentheses in the type designation will be deleted or replaced by a letter identifying the particular design for example: SG-12F/U or AN/URM-73C.

6.4 Group C inspection. Approval to ship may be withheld, at the discretion of the Government, pending the decision from the contracting officer on the adequacy of corrective action. (See 4.5.3.3)

6.5 Temperature conversion. Temperature conversion from fahrenheit to centigrade as listed in this specification are as follows:

$160^{\circ}F = 71^{\circ}C$

$125^{\circ}F = 51.5^{\circ}C$

$77^{\circ}F = 25^{\circ}C$

$40^{\circ}F = 4.44^{\circ}C$

$-4^{\circ}F = 20^{\circ}C$

$-65^{\circ}F = -53.9^{\circ}C$

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6.6 Verification inspection. Verification inspection by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

Custodians:

Army - EL

Preparing activity

ARMY - EL

Project Nr. 6625-A423

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. *Comments and the return of this form* will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions *submitted on* this form do *not constitute or imply authorization to waive any* portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

MIL-T-55687(EL), TEST SET, INSULATION BREAKDOWN AN/GSM-6( )

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

ESC-FM 1068-68

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